Practical no. 12

Aim – To interface an R307/R305 optical fingerprint sensor and a 1-channel relay module with a Raspberry Pi 4 and simulate a fingerprint-authenticated door-lock system.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. R307/R305 fingerprint sensor

- **Software Requirements**

1. Raspbian Stretch OS

1. Connect with Raspberry Pi's GPIO Pins.

- **Relay Module**

Device / Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO17	11

- **R307/R305 Fingerprint**

VCC	5V Pin 2	VCC
GND	GND Pin 6	GND
TX	GPIO15 / RXD, Pin 10	TX
RX	GPIO14 / TXD, Pin 8	RX

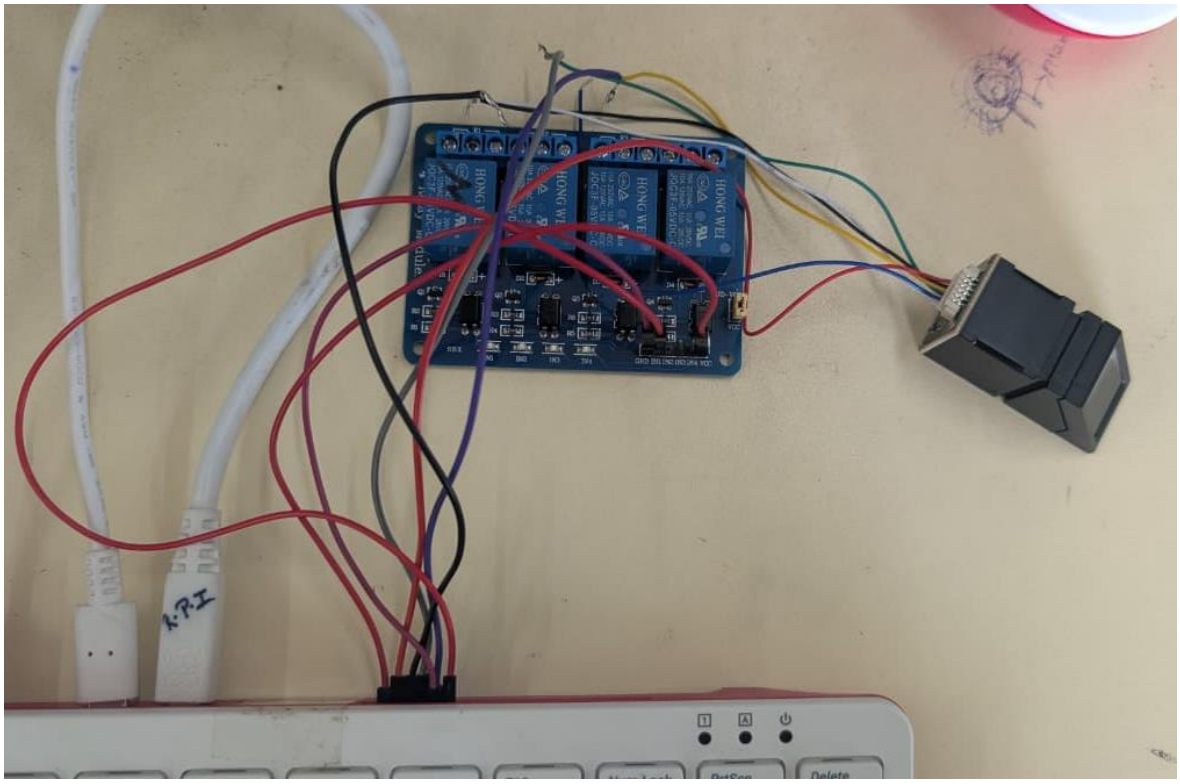
Software Installation
Step 1: Update Raspberry Pi packages
sudo apt update
sudo apt upgrade –y

Step 2: Install Python and serial support
sudo apt install -y python3 python3-pip python3-serial

Step 3: Install fingerprint library
pip3 install adafruit-circuitpython-fingerprint

Step 4: Enable Raspberry Pi UART
sudo raspi config
Go to Interface Options ------Serial Port
Login shell over serial: No
Serial port hardware: Yes
sudo reboot
After reboot, check the serial device:
ls -l /dev/serial0

If Raspberry Pi OS restricts system-wide pip installation
python3 -m venv fingerprint-env
source fingerprint-env/bin/activate
pip install pyserial adafruit-circuitpython-fingerprint RPi.GPIO



CODE:-

- **Fingerprint Enrollment**

```
import serial
import time
import adafruit_fingerprint
uart = serial.Serial(
    "/dev/serial0",
    baudrate=57600,
    timeout=1
)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)
finger_id = int(input("Enter fingerprint ID (1-127): "))
print("Place finger on sensor...")
while finger.get_image() != adafruit_fingerprint.OK:
    time.sleep(0.1)
print("Image captured")
if finger.image_2_tz(1) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()
print("Remove finger")
time.sleep(2)
print("Place the same finger again...")
while finger.get_image() != adafruit_fingerprint.OK:
    time.sleep(0.1)
if finger.image_2_tz(2) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()
if finger.create_model() != adafruit_fingerprint.OK:
    print("Fingerprints did not match")
    exit()
if finger.store_model(finger_id) == adafruit_fingerprint.OK:
    print("Fingerprint enrolled successfully")
    print("Stored ID:", finger_id)
else:
    print("Failed to store fingerprint")
```

- **Main Door Lock Verification Program**

```
import serial
import time
import RPi.GPIO as GPIO
import adafruit_fingerprint
GPIO.setwarnings(False)
# Change this GPIO if required
RELAY_PIN = 17
# Active-LOW relay
RELAY_ON = GPIO.LOW
RELAY_OFF = GPIO.HIGH
GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY_PIN, GPIO.OUT, initial=RELAY_OFF)

try:
    uart = serial.Serial(
        "/dev/serial0",
        baudrate=57600,
        timeout=1
    )
    finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)
    print("Fingerprint Door Lock System")
    print("Place Finger")
    while True:
        result = finger.get_image()
        if result == adafruit_fingerprint.OK:
            print("Processing...")
            # Convert image to template
            if finger.image_2_tz(1) != adafruit_fingerprint.OK:
                print("Fingerprint processing failed")
                time.sleep(1)
                continue
            # Search fingerprint database
            if finger.finger_search() == adafruit_fingerprint.OK:
                print("Access Granted")
                GPIO.output(RELAY_PIN, RELAY_ON)
                print("Door Unlocked")
                time.sleep(5)
                GPIO.output(RELAY_PIN, RELAY_OFF)
                print("Door Locked")
            else:
                print("Access Denied")
                GPIO.output(RELAY_PIN, RELAY_OFF)
            time.sleep(1)
            print("Place Finger")
        else:
            # No finger or another sensor response
            time.sleep(0.1)
```

```
except Exception as e:
    print("Communication error:", e)
finally:
    GPIO.output(RELAY_PIN, RELAY_OFF)
    GPIO.cleanup()
```

Run the Main Program

Run the program in thonny or IDLE

python3 enroll.py
python3 door_lock.py

Output –

Fingerprint enrollment o/p -

Enter fingerprint ID (1-127): 1
Place finger on sensor...
Image captured
Remove finger
Place the same finger again...
Fingerprint enrolled successfully
Stored ID: 1

Main Door Lock Verification Program o/p -

Fingerprint Door Lock System
Place Finger
Processing...
Access Granted
Door Unlocked
Door Locked
Place Finger